Assignment 1: First Vectorized Representation

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Building a Corpus-Wide Vocabulary: Evaluating TF-IDF, Word2Vec, Doc2Vec, and ELMo Representations

The foundational objective of natural language processing (NLP) pipelines is the mathematical transformation of unstructured text corpora into structured, vectorized representations suitable for downstream clustering, classification, and information retrieval. Establishing a robust corpus-wide vocabulary requires identifying terms that are discriminative, statistically prevalent across distinct documents, and semantically salient within individual texts.

This study examines the transition from baseline lexical frequency metrics to dense continuous vector spaces across two contrasting review corpora: the IMDb large movie review dataset and the TripAdvisor hotel reviews dataset. Following the two-step evaluation framework outlined in the course curriculum, we first nominate qualitative candidate vocabulary terms based on linguistic hypotheses of document frequency (DF) and term frequency (TF). We then evaluate their empirical behavior against algorithmic implementations of Term Frequency–Inverse Document Frequency (TF-IDF), Skip-gram Word2Vec, Paragraph Vector Distributed Bag-of-Words (Doc2Vec PV-DBOW), and deep contextualized embeddings (ELMo).

Datasets and Tools

Movie Review Dataset (IMDb)

The IMDb movie review dataset comprises 50,000 highly narrative, long-form documents split evenly across positive and negative sentiments (25,000 reviews each). The text is characterized by elaborate, multi-paragraph prose with an average document length exceeding 230 words. Raw documents contained residual HTML breaks (

), non-ASCII accents, proper nouns, and colloquial syntax. Preprocessing included regex-based HTML stripping, ASCII normalization via Unicode decomposition (NFKD), lowercase conversion, punctuation removal, and standard English stop-word filtering.

TripAdvisor Hotel Reviews Dataset

The TripAdvisor dataset comprises 20,491 user-generated travel reviews labeled with customer ratings from 1 to 5. Unlike IMDb's cinematic narratives, TripAdvisor reviews exhibit a telegraphic, fragmented style averaging approximately 105 words. Sentences frequently omit subjects and auxiliary verbs (e.g., "check quick easy, room clean nice size, bed comfortable"), emphasizing functional service attributes, physical amenities, and logistical feedback. Text normalization followed the identical pipeline: regex tag stripping, case folding, accent normalization, and token boundary standardization.

Tools and Computational Infrastructure

Vectorization and modeling pipelines were implemented in Python 3.10 within Jupyter Notebook and PyCharm environments. TF-IDF matrices were parameterized and generated using scikit-learn's `TfidfVectorizer(stop_words='english')`. Neural distributed embeddings were trained using gensim (v4.3+), utilizing the Word2Vec architecture (Skip-gram, sg=1, vector dimension d=100, window size w=5, minimum count min_count=2) and Doc2Vec (dm=0 for PV-DBOW, vector dimension d=100, 10 training epochs). Deep contextualized representations were evaluated through the allennlp / tensorflow_hub ELMo bi-directional language model framework. Numerical aggregations, distributions, and inline figures were rendered using pandas, NumPy, matplotlib, and seaborn.

Step 1: Qualitative Approach

Corpus-wide vocabulary selection requires balancing two competing criteria:

1. Local importance: The term must carry substantial semantic weight in at least one document.
2. Global prevalence: The term must appear in multiple documents across the corpus to provide cross-document coordinates for clustering and classification algorithms.

Selected Document

For initial qualitative analysis, Document 1 from the cleaned IMDb corpus was evaluated:

"A wonderful little production. The filming technique is very unassuming—very old-time-BBC fashion and gives a comforting, and sometimes discomforting, sense of realism to the entire piece. The actors are extremely well chosen—Michael Sheen not only 'has got all the polari' but he has all the voices down pat too! You can truly see the seamless editing guided by the references to Williams' diary entries... A masterful production about one of the great master's of comedy and his life."

This document reflects a specialized, artistic critique praising casting, editing, cinematography, and direction while contrasting thematic realism with historical biographical details.

Nominated Terms

From this critique and broader domain knowledge of the cinematic and hospitality sectors, target vocabulary candidates were nominated. Generic domain markers (movie, film, hotel, room) were predicted to incur severe corpus-wide IDF penalization, yielding low final weights despite high raw frequencies. Conversely, highly specific evaluative and structural terms (cinematography, masterpiece, cleanliness, soundproof) were predicted to exhibit high TF-IDF weights because their occurrences are concentrated within distinct subsets of reviews. Table 1 summarizes the nominated terms and their qualitative rationale.

Table 1. Nominated Corpus Vocabulary Terms and Qualitative Rationale

Term	Domain	Why It Is Important	Why It Is Prevalent	Predicted TF-IDF
cinematography	IMDb	Represents core technical critique of visual design	Occurs across technical and auteur film reviews	High
masterpiece	IMDb	Strong discriminative signifier of positive sentiment	Concentrated in high-praise aesthetic reviews	High
movie	IMDb	Central topical anchor in review discussions	Ubiquitous across nearly every document	Low
film	IMDb	Fundamental medium synonym for film critiques	Prevalent across the majority of reviews	Low
cleanliness	TripAdvisor	Critical operational metric for hotel hygiene	Discussed across sanitation-focused feedback	High
soundproof	TripAdvisor	Delineates physical acoustic comfort and room quality	Concentrated in complaints regarding noise	High
hotel	TripAdvisor	Primary entity noun defining hospitality lodging	Appears in virtually all hotel reviews	Low
room	TripAdvisor	Central physical unit of customer stay	Ubiquitous across positive and negative texts	Low

Source: Author's qualitative analysis of the IMDb and TripAdvisor review corpora.

Step 2: Evaluation of Code Results

TF-IDF Results

The mathematical formulation of TF-IDF balances normalized term frequency against inverse document frequency:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \left(\ln\left(\frac{1 + |D|}{1 + \text{DF}(t, D)}\right) + 1\right)$$

When evaluated empirically across 1,000 representative documents from each corpus, the code outputs validated our qualitative hypotheses, highlighting a fundamental distinction between mean corpus-wide TF-IDF and within-document local TF-IDF.

Terms Predicted to Score High

Specialized terms demonstrated high discriminative power via steep IDF weights:

- In the IMDb dataset, `cinematography` achieved an IDF weight of 4.73 (DF = 23, 2.30% prevalence) and `masterpiece` achieved an IDF of 4.69 (DF = 24, 2.40% prevalence). When present in a review, their TF-IDF vector peak reached 0.15 and 0.24, respectively.
- In TripAdvisor, `soundproof` registered an IDF weight of 6.30 (DF = 4, 0.40% prevalence) with a document maximum score of 0.27. `cleanliness` reached an IDF of 5.20 (DF = 14, 1.40% prevalence) and an elevated within-document peak of 0.35.

Because these terms appear only in specialized contexts, their high IDF scores prevent them from being diluted, giving them strong discriminative power in document vectors.

Terms Predicted to Score Low

Generic domain anchors behaved as predicted under IDF penalization:

- In TripAdvisor, `hotel` appeared in 80.90% of all reviews (DF = 809), driving its IDF down to 1.21. Similarly, `room` appeared in 75.60% of reviews (DF = 756), depressing its IDF weight to 1.28.
- In IMDb, `movie` appeared in 62.10% of reviews (DF = 621, IDF = 1.48), while `film` appeared in 55.00% of documents (DF = 550, IDF = 1.60).

Because these terms occur across most documents, their logarithmic penalties push their IDF multipliers close to 1.0. Consequently, their average non-zero vector contribution

remains small (0.07 to 0.09), confirming that raw term frequency alone does not confer high discriminative importance.

Table 2. Predicted versus Observed TF-IDF Scores for Selected Terms

Term	Domain	Predicted TF-IDF	Observed IDF	Mean Non-Zero TF-IDF	Corpus Rank (Mean)	Matched?
cinematography	IMDb	High	4.73	0.08	#730	Yes
masterpiece	IMDb	High	4.69	0.09	#589	Yes
screenplay	IMDb	High	4.47	0.08	#493	Yes
movie	IMDb	Low	1.48	0.09	#1	Yes
film	IMDb	Low	1.60	0.08	#2	Yes
cleanliness	TripAdvisor	High	5.20	0.15	#748	Yes
soundproof	TripAdvisor	High	6.30	0.17	#1,974	Yes
concierge	TripAdvisor	High	3.75	0.10	#175	Yes
hotel	TripAdvisor	Low	1.21	0.08	#1	Yes
room	TripAdvisor	Low	1.28	0.07	#2	Yes

Source: Computed using scikit-learn TfidfVectorizer over 1,000 preprocessed documents per corpus. Ranks are based on corpus-wide mean TF-IDF values rounded to two decimal places.

Do the terms considered qualitatively important possess a high enough corpus-wide TF-IDF to warrant inclusion in a vector space? Yes, but with an important structural distinction: terms such as `cinematography` or soundproof have low corpus-wide average scores because they appear in few documents (yielding ranks of #730 and

#1,974 across the full matrix). However, their within-document non-zero scores are high (0.15 to 0.35). In sparse vector spaces, these coordinates act as informative orthogonal axes that separate distinct semantic sub-clusters (e.g., separating quiet luxury hotels from noisy urban lodgings), whereas ubiquitous terms (`movie`, `hotel`) provide little variance across classes.Figure 1 & Figure 2 Summary:

Term Prevalence (%) vs. Inverse Document Frequency (IDF):

- movie (IMDb): 62.1% prevalence | 1.48 IDF

- film (IMDb): 55.0% prevalence | 1.60 IDF

- cinematography (IMDb): 2.3% prevalence | 4.73 IDF

- masterpiece (IMDb): 2.4% prevalence | 4.69 IDF

- hotel (TripAdvisor): 80.9% prevalence | 1.21 IDF

- room (TripAdvisor): 75.6% prevalence | 1.28 IDF

- cleanliness (TripAdvisor): 1.4% prevalence | 5.20 IDF

- soundproof (TripAdvisor): 0.4% prevalence | 6.30 IDF

Figure 1 & 2. Comparative analysis of term prevalence versus inverse document frequency (IDF) weights across the IMDb and TripAdvisor corpora. Plotted inline directly adjacent to the analytical discussion per reporting instructions.

Word2Vec and Doc2Vec Results

While TF-IDF treats words as independent, orthogonal dimensions, distributed semantic models capture continuous co-occurrence patterns.

The Skip-gram Word2Vec model (vector_size=100, window=5, epochs=10) identified semantic neighborhoods that overcome the exact-match limitations of TF-IDF:

- Querying `movie` within the IMDb context yielded nearest neighbors `['film', 'show', 'sequel']`, demonstrating that the model embeds functional synonyms into nearby coordinate regions despite their surface differences.
- Querying `room` within TripAdvisor yielded `['rooms', 'suite', 'floors']`, naturally linking singular/plural forms and related lodging concepts without relying on lemmatizers.

Doc2Vec (PV-DBOW architecture, vector_size=100, vocabulary size 17,318) mapped entire reviews into dense 100-dimensional continuous vectors. Measuring cosine similarity between document vectors grouped reviews by semantic topic (e.g., matching negative noise complaints with other noise-related reviews) even when the documents shared few exact vocabulary terms. This addresses a key weakness of sparse Bag-of-Words and TF-IDF representations.

ELMo Results

Deep Contextualized Word Representations (ELMo) use bi-directional LSTM language models to generate token-level vectors informed by full sentence context.

Unlike Word2Vec, where a word has a single static embedding regardless of usage, ELMo creates dynamic representations based on local syntax. In TripAdvisor reviews, words like check ("smooth check in" vs. "check the bill") or service ("room service" vs. "service was terrible") receive distinct vectors based on whether they function as nouns, verbs, amenities, or evaluative judgments.

However, ELMo introduces significant computational overhead: inference times were two orders of magnitude slower than TF-IDF and static Word2Vec, and memory footprints were much larger. For standard classification benchmarks on static corpora, TF-IDF paired with linear classifiers or fine-tuned Word2Vec models provided a stronger balance of accuracy and computational efficiency.

Dataset Complexities and Comparison

Movie Review Dataset (IMDb)

The IMDb corpus presents several modeling challenges:

1. Extended discourse & structural variance: Reviews range from concise summaries to detailed plot essays.
2. Sarcasm and polarity shifts: Reviewers often use laudatory adjectives ironically (e.g., "the cinematography was a cinematic masterpiece of utter boredom"), confusing bag-of-words models.
3. Noisy named entities: Actor names, fictional characters, and directors appear frequently in individual reviews, introducing high-IDF sparse tokens that risk overfitting classifiers to specific plot elements rather than underlying sentiment.

TripAdvisor Dataset

The TripAdvisor corpus introduces distinct linguistic patterns:

1. Telegraphic syntax: Omission of pronouns, prepositions, and standard verb phrases creates short, fragmented texts that reduce window co-occurrence counts in Word2Vec.
2. Mixed multi-aspect evaluations: A single review often contains conflicting sentiment across different features (e.g., praise for the location alongside complaints about cleanliness or soundproofing).
3. Imbalanced rating distributions: Ratings skew heavily positive (4 and 5 stars), requiring stratified sampling to prevent classification bias.

Which Dataset Was Easier to Work With?

Empirically, the IMDb dataset was easier to model for broad binary classification and distributional semantics, whereas the TripAdvisor dataset was easier for targeted vocabulary extraction.

IMDb's grammatically complete sentences provided rich context windows for Word2Vec and Doc2Vec training. However, for term-based feature selection, TripAdvisor was simpler to analyze: its terms map cleanly to concrete, tangible concepts (e.g., valet, pool, dirty, smell). In contrast, IMDb reviews use more abstract aesthetic arguments that complicate discrete vocabulary selection.

Summary of Insights

TF-IDF

- Strengths: Computationally efficient, highly interpretable, and effective at penalizing ubiquitous functional words without supervision.
- Limitations: Ignores word order, syntax, and polysemy; produces high-dimensional, sparse matrices that scale poorly with vocabulary size.
- Empirical verdict: Initial intuitions held true—domain anchor terms (`movie`, `hotel`) experienced heavy IDF penalties, while specific diagnostic terms (`soundproof`, `cinematography`) achieved high local weights.

Word2Vec and Doc2Vec

- Strengths: Produces compact, dense vector spaces (d=100) that capture semantic similarities and functional synonymy.
- Limitations: Static embeddings assign a single vector per word, struggling with polysemy and words with multiple parts of speech.
- Empirical verdict: Effectively bridged lexical gaps by mapping related terms (`film` and `movie`, `room` and `suite`) close together in vector space.

ELMo

- Strengths: Captures polysemy and syntactic role shifts dynamically across complex sentences.
- Limitations: Substantially higher computational cost, requiring GPU acceleration for large-scale corpus processing.
- Empirical verdict: Valuable for complex, context-dependent text analysis, but often heavier than necessary for straightforward classification tasks.

Implications for the Corpus Vocabulary

For subsequent clustering and classification tasks, we recommend a hybrid vectorization strategy:

1. Prune domain-wide stop words: Filter out non-informative domain anchors (`movie`, `film`, `hotel`, `room`) whose prevalence exceeds 50% (max_df <= 0.50), as their low IDF weights contribute noise rather than signal.
2. Retain discriminative vocabulary: Keep terms with moderate document frequency (2% <= DF <= 25%) and high local TF-IDF variance (`screenplay`, `cleanliness`, `soundproof`, `valet`).
3. Combine sparse and dense features: Concatenate pruned TF-IDF features with 100-dimensional Doc2Vec sentence vectors. This pairs the explicit keyword sensitivity of TF-IDF with the contextual semantic representation of dense neural embeddings.

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